Ch 7.1-7.2: Polynomial regression and Step Functions

Lecture 20 - CMSE 381

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Announcements

Last time:

- Exam

This lecture:

- 7.1 Polynomial regression
- 7.2 Step functions

Announcements:

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Section 1

2 times ago ~~Last time~~

High-Dimensional Data

Low-Dimensions

$n \gg p$

- Low here means p is low, or at least small relative to n
- Can do all the stuff we've talked about so far

High-Dimensions

$n \ll p$

- Issues show up even if p is close to or slightly smaller than n
- Classical approaches not appropriate since lots of overfitting

What to do about it?

Be less flexible....

Key points

- regularization or shrinkage plays a key role in high-dimensional problems,
- appropriate tuning parameter selection is crucial for good predictive performance, and
- the test error tends to increase as the dimensionality of the problem increases, unless the additional features are truly associated with the response.

- Curse of dimensionality
- Report results on an independent test set, or cross-validation errors.

Section 2

Polynomial Regression

Want more flexibility

Polynomial regression

Replace linear model

$$y_i = \beta_0 + \beta_1 x_1 + \varepsilon_i$$

$p = 1$

with

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

pass matrix with powers of the single input variable as other columns

Change d to suit

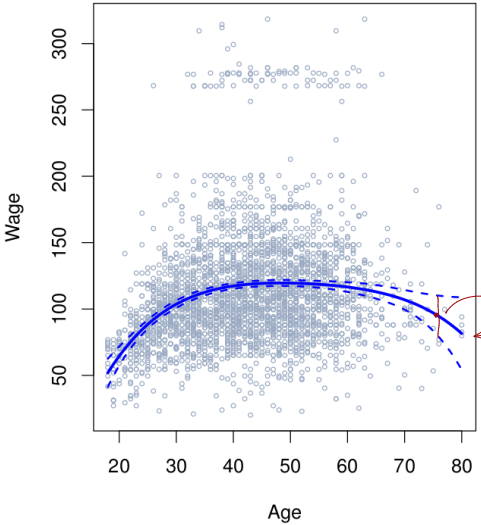
tend to have $d \leq 3$

Coding bit

$d = p$

$$\texttt{wage} = \beta_0 + \beta_1 \texttt{age} + \beta_2 \texttt{age}^2 + \cdots + \beta_p \texttt{age}^p + \varepsilon.$$

Example with wage data



know Var($\hat{f}(x_0)$)

Draw confidence intervals

deg 4 polynomial

(same as in notebook

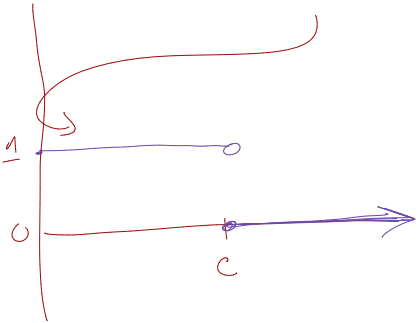
Section 3

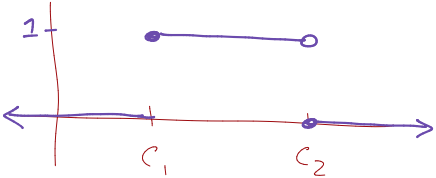
Step function

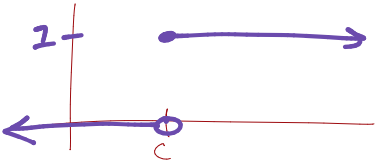
Step functions

$X \in [c_1, c_2)$

$\mathrm{I}(X < c) \qquad \mathrm{I}(c_1 \leq X < c_2) \qquad \mathrm{I}(c \leq X)$



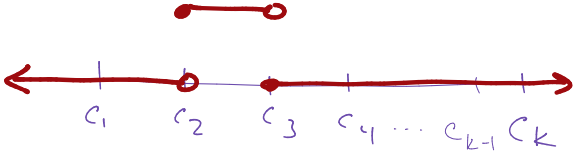




More on step function setup

Fix values $c_1, ___, c_K$ ← Knots

$$
\begin{aligned}
C_0(X) &= I(X < c_1), \\
C_1(X) &= I(c_1 \le X < c_2), \\
C_2(X) &= I(c_2 \le X < c_3), \\
&\vdots \\
C_{K-1}(X) &= I(c_{K-1} \le X < c_K), \\
C_K(X) &= I(c_K \le X),
\end{aligned}
$$



Example

Given knots $c_1 = 3$, $c_2 = 5$, $c_3 = 7$, determine the entries in the columns for $C_i(X)$ in the below matrix.

	$(-\infty, 3)$	$[3, 5)$	$[5, 7)$	$[7, \infty)$
X	$C_0(X)$	$C_1(X)$	$C_2(X)$	$C_3(X)$
1	1	0	0	0
7	0	0	0	1
3	0	1	0	0
5	0	0	1	0
4	0	1	0	0
9	0	0	0	1

- Only one one 1 in any row
- $\sum_{j=0}^{k} C_j(x) = 1$

Step function: Learned model

Drops C_0 Because complete info in other columns

$$y_i = \beta_0 + \underbrace{\beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i)} + \varepsilon_i$$

* Break up X values into bins and learn coeff. for that bin
* If data $X < c_1$, P

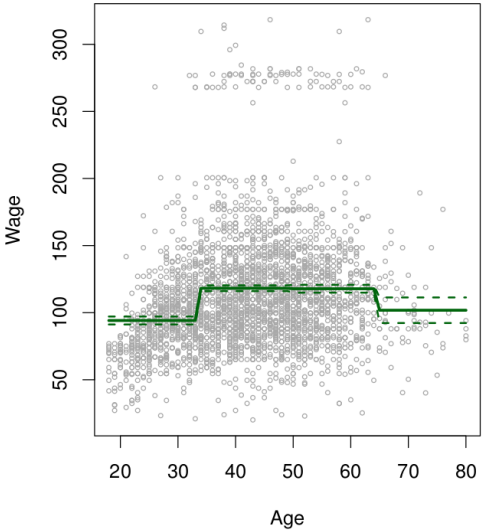
So β_0 = average y-value for $X < c_1$

Other coeff can be viewed increase / decrease in response of Y for bin $c_i \leq X \leq c_{i+1}$ relative to

Coding bit

Back to the `wage` data set

Step function example



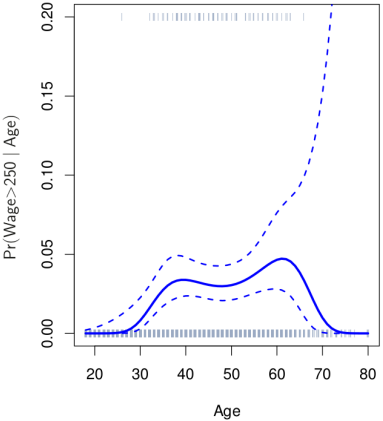
Section 4

Classification versions

Next time!

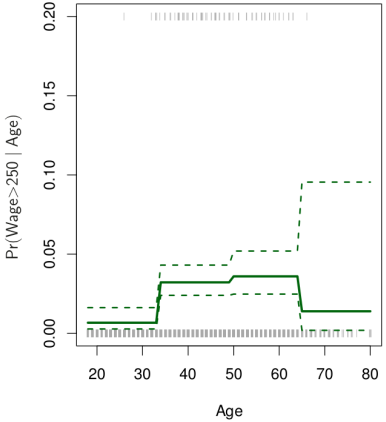
Classification version

$$\Pr(y_i > 250 \mid x_i) = \frac{\exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}$$



Step function classification example

$$\Pr(y_i > 250 \mid x_i) = \frac{\exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}{1 + \exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}$$



A few more comments on step functions

Basis Functions Setup

Polynomial and piecewise-constant regression models are special cases of a *basis function* approach.

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

Next time

20	F	Nov 4	Polynomial & Step Functions.	7.1,7.2	
21	M	Nov 7	Basis functions, Regression Splines	7.3,7.4	
22	W	Nov 9	Smoothing Splines; Local regression; GAMs	7.5-7.7	
23	F	Nov 11	Decision Trees	8.1	HW #7 Due
24	M	Nov 14	Ensemble methods	8.2	
25	W	Nov 16	Maximal Margin Classifier	9.1	
26	F	Nov 18	SVC	9.2	HW #8 Due
27	M	Nov 21	SVM	9.3, 9.4, 9.5	
28	W	Nov 23	Single layer NN	10.1	
	F	Nov 25	No class - Thanksgiving		
29	M	Nov 28	Multi Layer NN	10.2	HW #9 Due
30	W	Nov 30	CNN	10.3	
31	F	Dec 2	Unsupervised Learning & Clustering	12.1, 12.4	
32	M	Dec 5	More Clustering	12.4	HW #10 Due
	W	Dec 7	Review		
	F	Dec 9	Midterm #3	Bring your cheat sheet and a non-internet-connected calculator	